
Waste & By-product Handling SOP — Bioethanol Plant (excerpt)

Supplied grounding document for capstone groups (e.g. a waste-reduction agent) that don't have their own. Covers the main waste/by-product streams around D-201.

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1. Purpose & scope

Defines normal handling, quality checks, and disposal routing for the plant's main waste and by-product streams: stillage (bottoms), off-spec distillate, and spent filter/strainer solids. Applies to all shift operators.

2. Streams covered

Stream	Source	Normal routing
Stillage (bottoms)	D-201 reboiler sump	To evaporation/DDGS recovery for animal feed by-product
Off-spec distillate	AT-207 sustained low-purity divert (SOP-D201-014 §6)	Slops tank, reprocessed through D-201 on next campaign
Strainer/filter solids	P-450 suction strainer, feed filters	Solid waste skip, licensed contractor collection
Spent process water	Wash-downs, sample points	To site water treatment plant (see Water Treatment SOP) — never to storm drain

3. Quality checks before routing

- **Stillage:** confirm bottoms temperature was within 98–107 °C normal range (SOP-D201-014 §3) before diverting to DDGS recovery; an out-of-range batch should be flagged for engineering review, not routed automatically.
- **Off-spec distillate (slops):** log the AT-207 reading and duration of the low-purity event triggering the divert. Slops above 80% v/v EtOH can usually be reprocessed directly; below that, blend ratio needs engineering sign-off.
- **Solids:** inspect for oil/chemical contamination before sending to the general skip — contaminated solids must go to hazardous waste, not general waste.

4. Minimising waste generation

1. **Reduce off-spec divert frequency** — most low-purity diverts trace back to reflux/feed upsets covered in SOP-D201-014 §8 (troubleshooting). Fixing the root cause reduces slops volume more than any downstream handling change.
2. **Strainer fouling** — rising pressure drop across the P-450 suction strainer increases both energy use and solids waste from more frequent cleaning; trend it rather than reacting to a single high reading.
3. **Batch consistency** — stillage quality (and its value as DDGS by-product) is best when the column runs steadily within its normal operating envelope; frequent startups/shutdowns degrade by-product value (see Startup/Shutdown SOP).

5. Escalation & human sign-off

- Any waste stream that fails the quality checks in §3 (contaminated solids, out-of-range stillage, low-purity slops below reprocessing threshold) must be flagged to a shift engineer for a routing decision — do not route automatically.
- Disposal contractor changes, hazardous-waste manifests, and any discharge to water require environmental/compliance sign-off, not operator or assistant action (see Effluent Compliance Standard).